

Visual Analytics of Housing Attributes and Pricing in the Larsen & Toubro Realty Dataset

Introduction:

The Indian real estate sector has witnessed rapid growth, fueled by urbanization, rising incomes, and changing lifestyle preferences. Among its leading developers, Larsen & Toubro (L&T) Realty stands out with a diverse portfolio of residential projects across major and emerging cities. Understanding how factors like location, property features, and connectivity influence prices is vital for developers, policymakers, and investors alike.

With the rise of large-scale housing data, data visualization offers an effective way to uncover market patterns and pricing dynamics. The L&T Realty dataset used in this study contains detailed information on property size, type, number of rooms, furnishing, amenities, distance to metro, and price. Through systematic visual analysis, this project aims to identify trends in housing markets, assess affordability, and explore how infrastructure impacts property values.

Ultimately, the project seeks to apply visualization techniques for real estate analysis and derive actionable insights from the data—supporting informed decision-making and a deeper understanding of urban housing patterns.

Data Description:

The dataset titled *L&T Realty Housing Data* contains information on **100,000 residential properties** across various Indian cities.

It includes **14 variables** describing property details, location factors, and prices.

Each record represents a **unique property listing** with both numerical and categorical features.

The variables are described as follows:

- **Property ID:** Unique identifier for each property.
- **Location:** City or area where the property is located.
- **Property Type:** Indicates whether the property is an apartment, villa, duplex, or independent house.
- **Size (sqft):** Built-up area of the property measured in square feet.
- **Number of Bedrooms:** Total number of bedrooms in the property.
- **Number of Bathrooms:** Total number of bathrooms available.
- **Floor Number:** The specific floor where the property is located.
- **Total Floors:** Total number of floors in the building.
- **Age (years):** Age of the property in years.

- **Furnishing:** Furnishing status—fully furnished, semi-furnished, or unfurnished.
- **Amenities:** List of facilities such as gym, swimming pool, parking, garden, and security.
- **Distance to Metro (km):** Distance from the property to the nearest metro station in kilometres.
- **View Score:** Rating (1–10) representing the quality of the property’s view.
- **Price:** Market price of the property in Indian Rupees.

The dataset is **complete and free from missing values**, ensuring reliability for analysis.

It captures both **physical** and **locational** attributes, making it suitable for **visualization**, **trend analysis**, and **price prediction studies** in the real estate domain.

EDA (Exploratory Data Analysis):

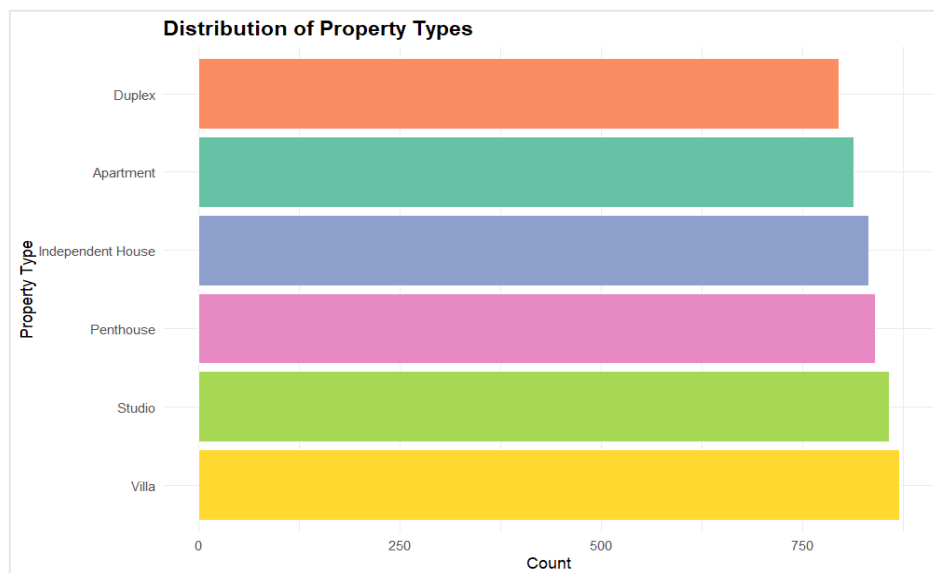


Fig 1: Counts of different property types

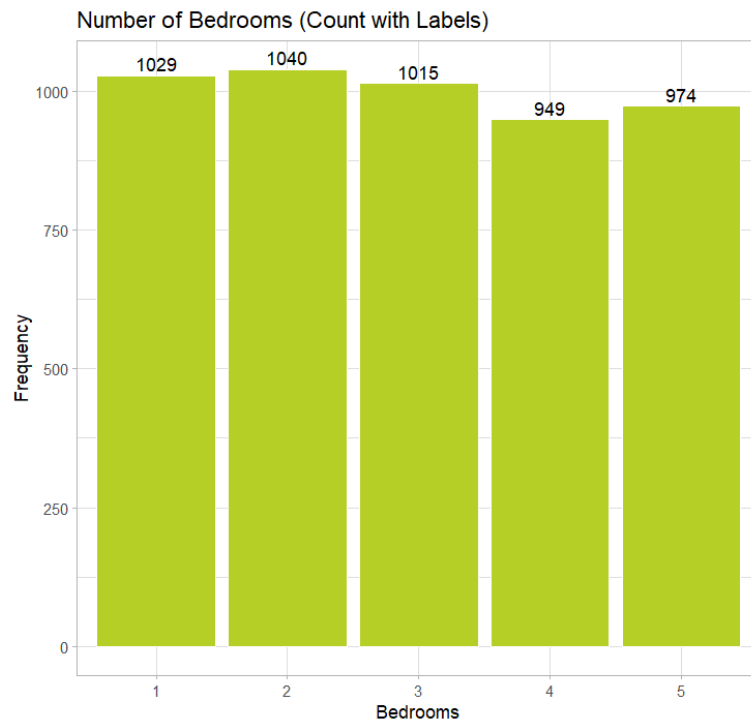


Fig 2: Count of number of bedrooms across different properties

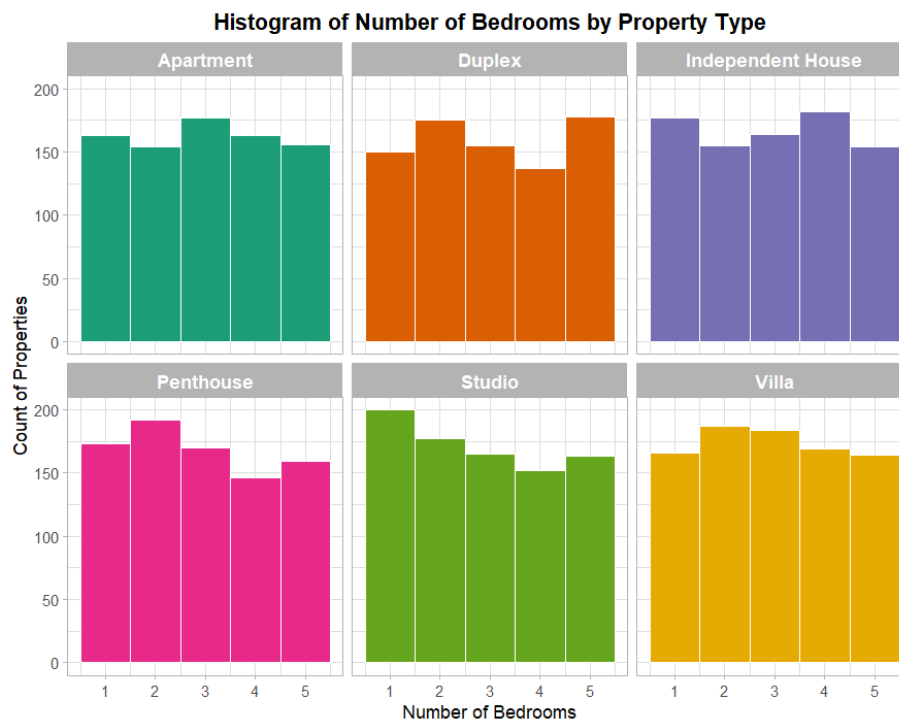


Fig 3: Number of bedrooms across different types of properties

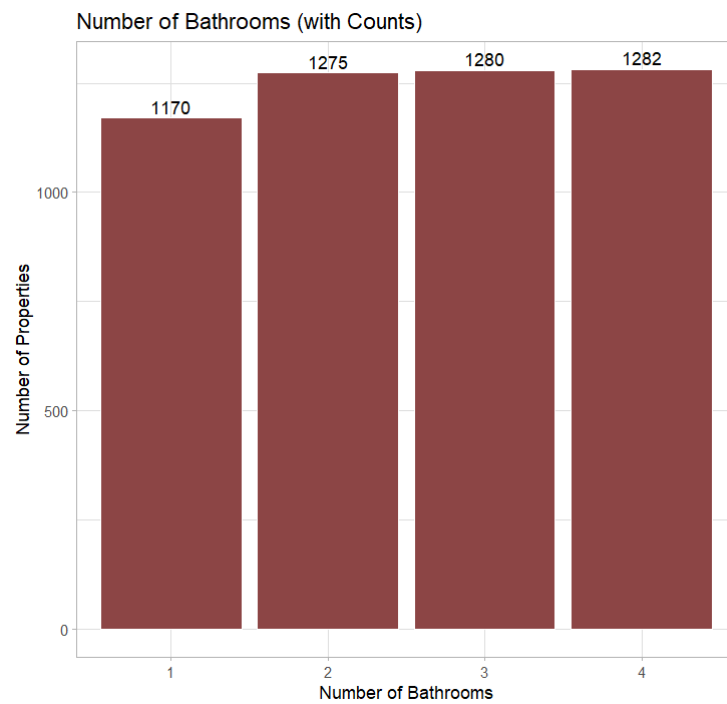


Fig 4: Number of bathrooms across different properties

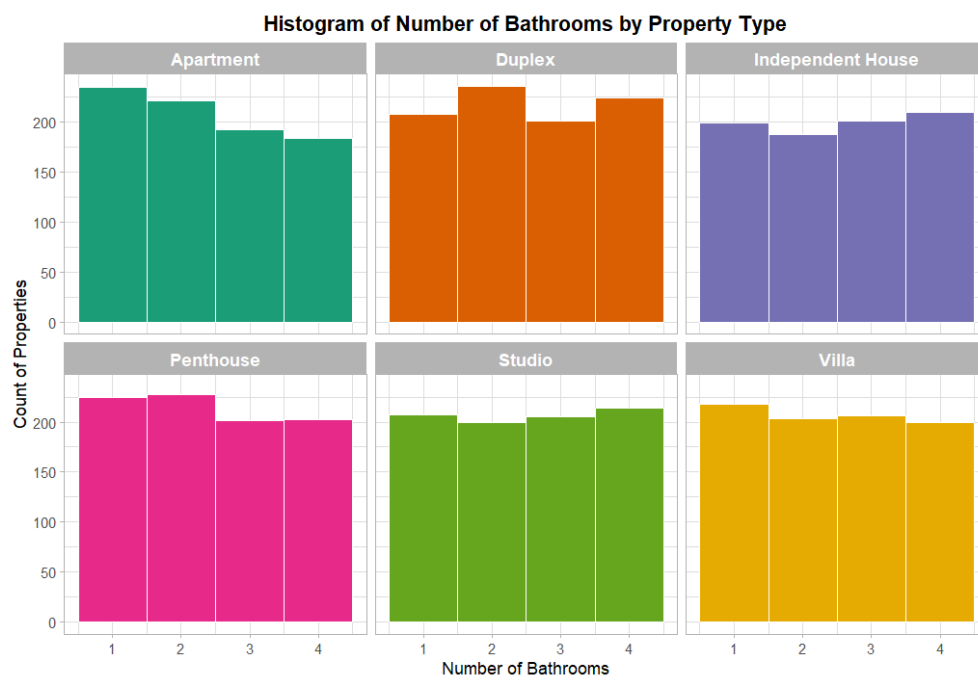


Fig 5: Number of bathrooms across different kinds of properties

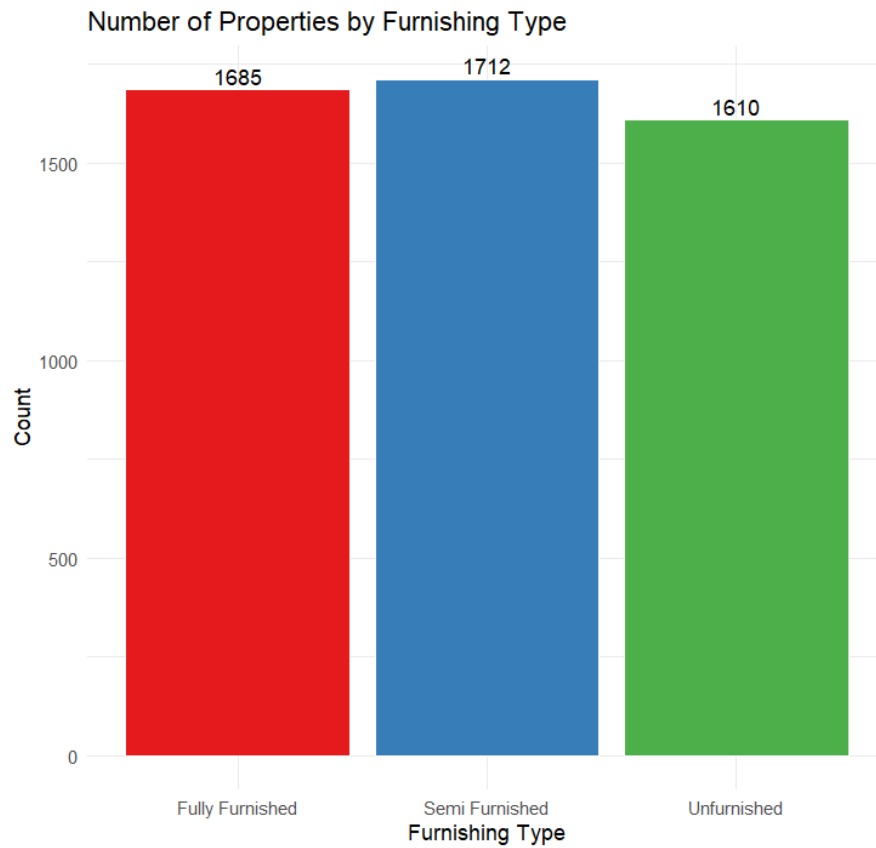


Fig 6: Bar chart of different furnishing type of the properties

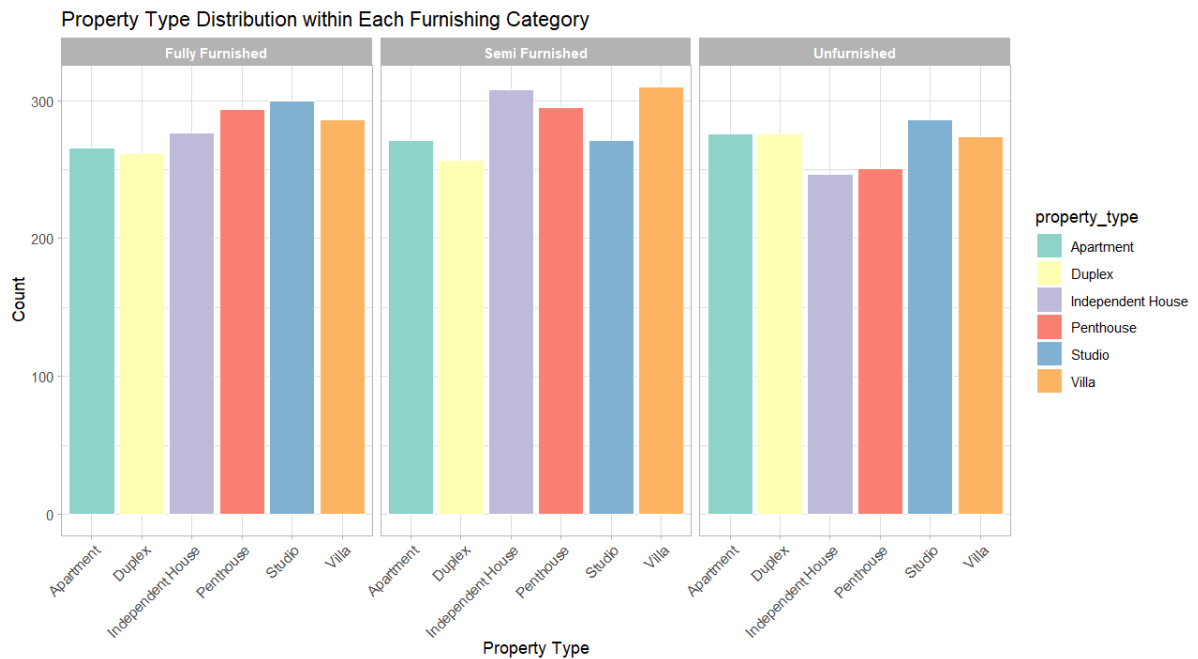


Fig 7: Furnishing types across different types of properties

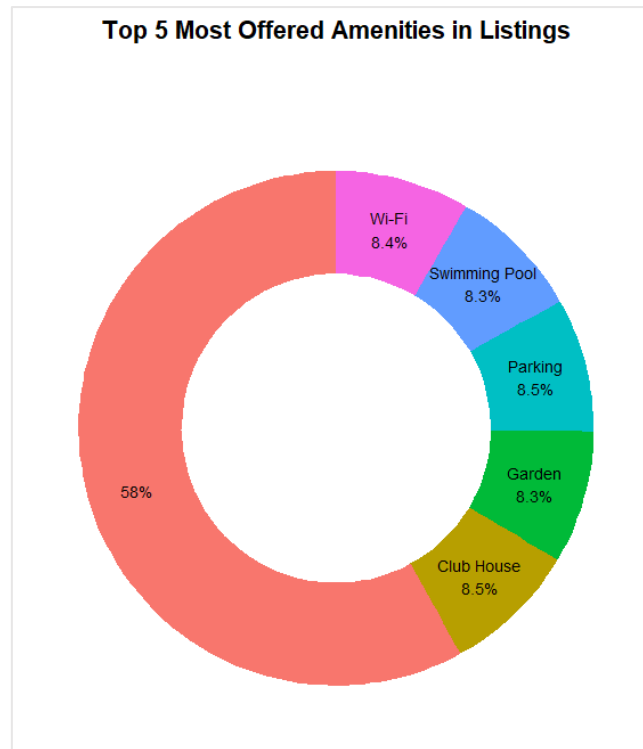


Fig 8: Donut chart of top 5 amenities provided to the properties

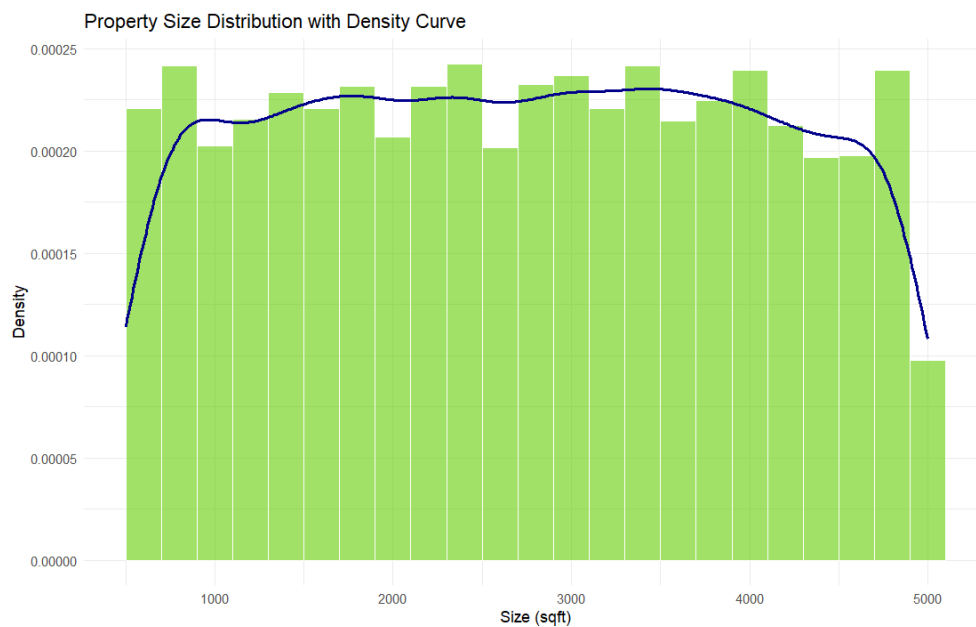


Fig 9: Histogram with density curve showing distribution of the property sizes



Fig 10: Density plots of size(sqft) for different types of properties

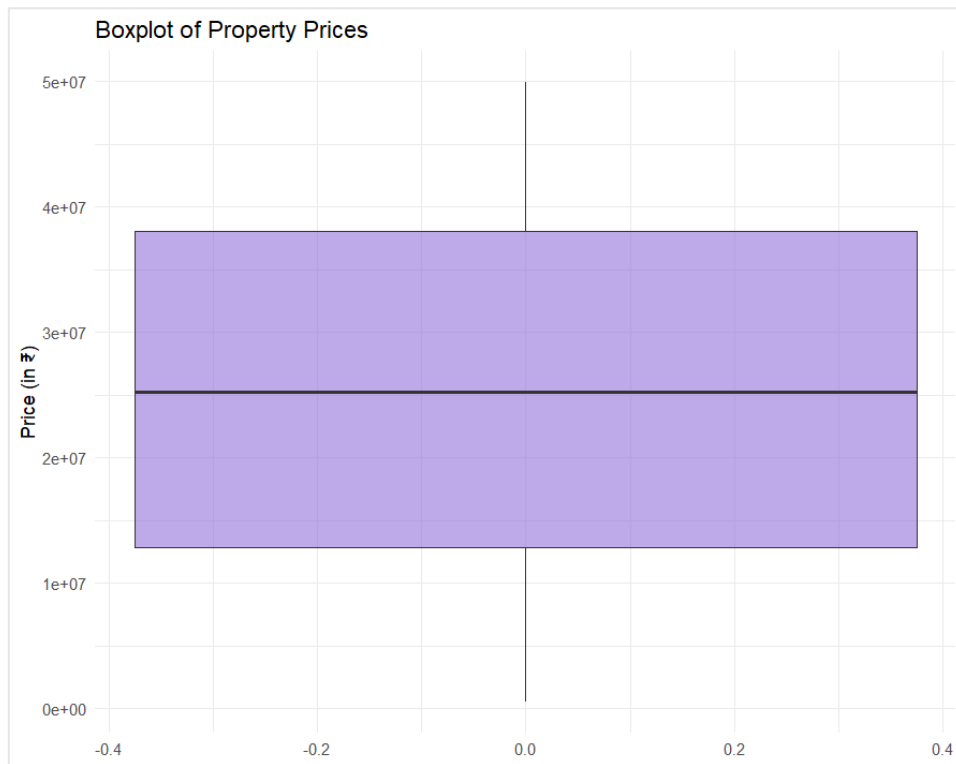


Fig 11: Boxplot for price of the properties



Fig 12: Histogram of log transformed house prices



Fig 13: Line graph of average price across different property types

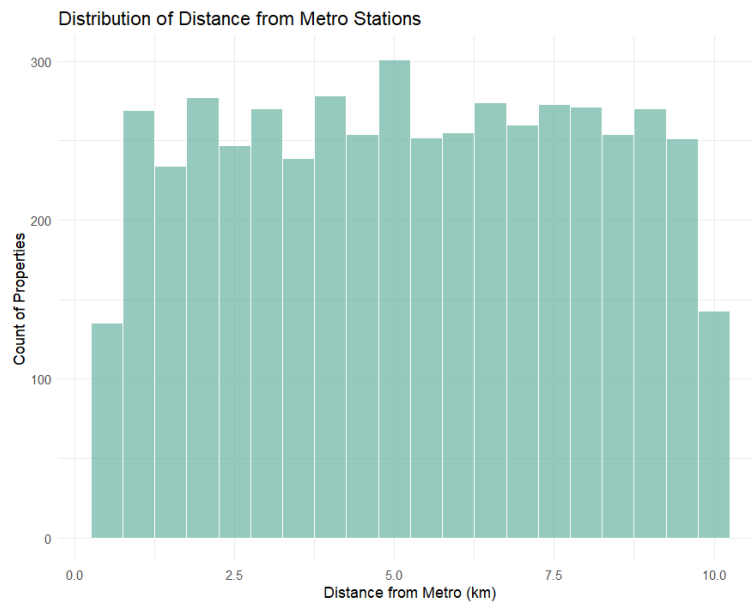


Fig 14: Histogram of distance of metro from the properties

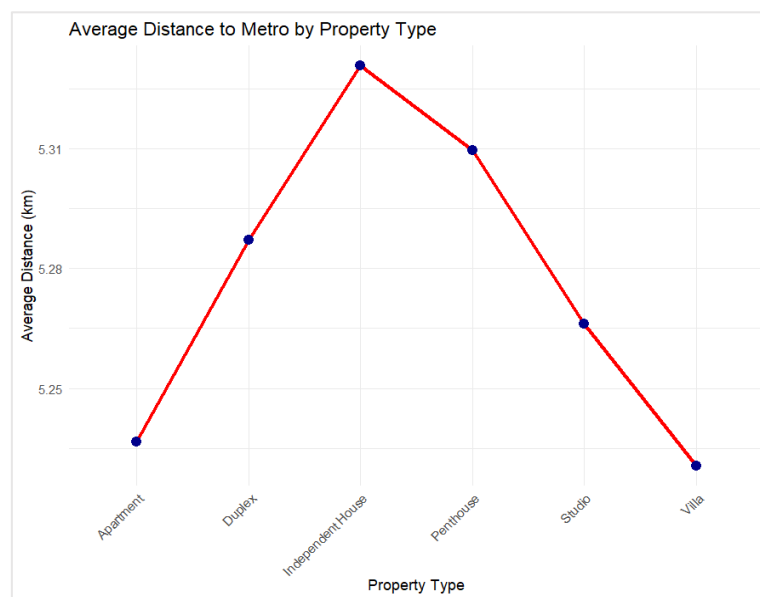


Fig 15: Line graph depicting average distance of metro for various kinds of properties

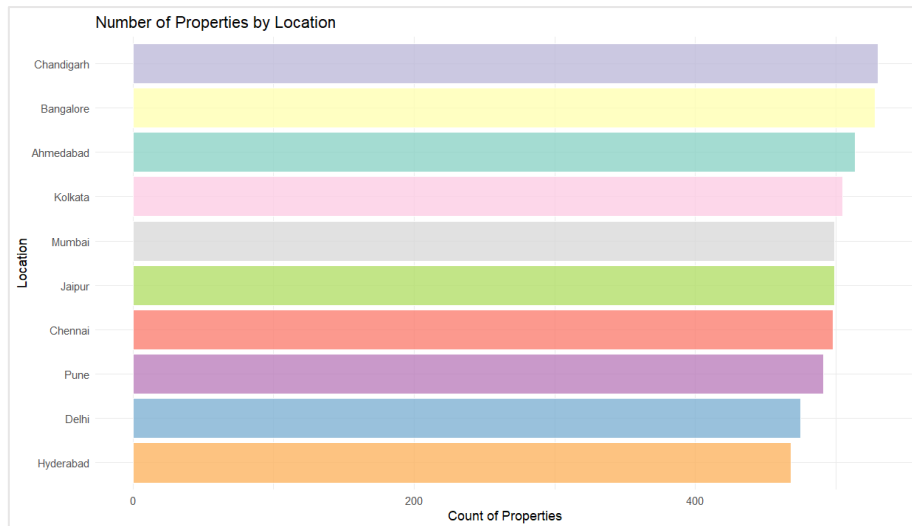


Fig 16: Number of properties built in different locations

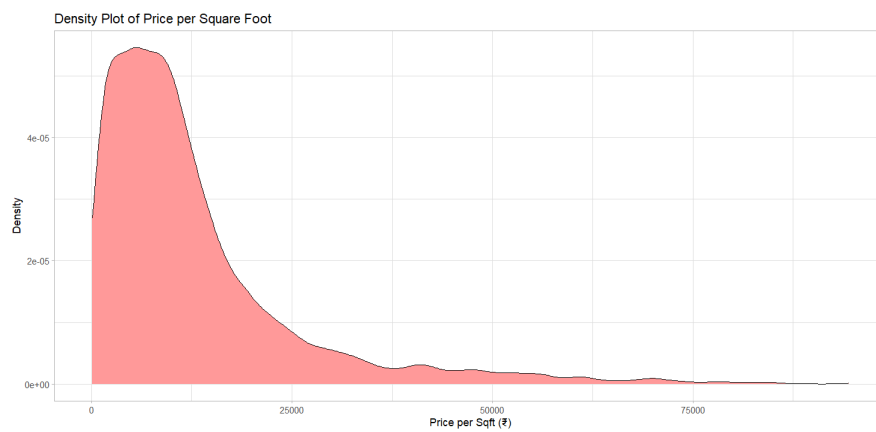


Fig 17: Density plot for price per square foot of the properties

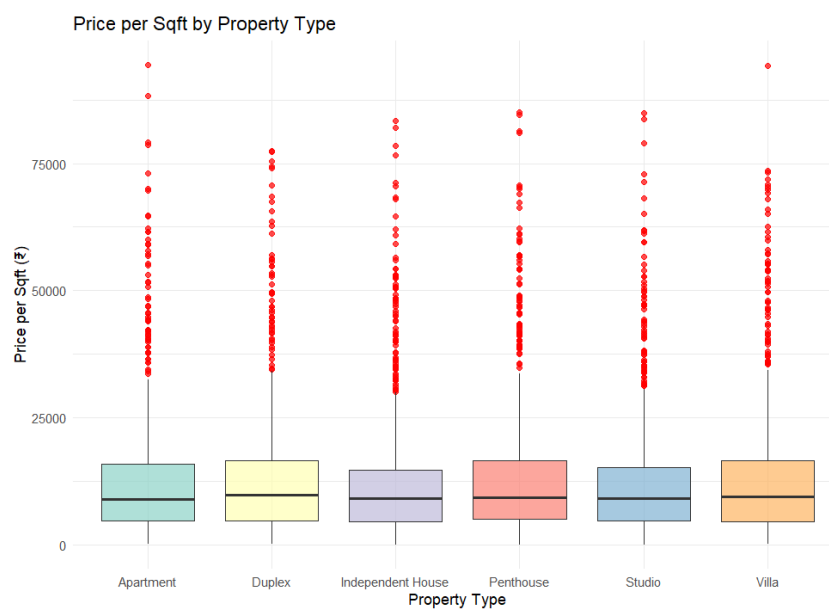


Fig 18: Boxplot of price per square foot for different property types

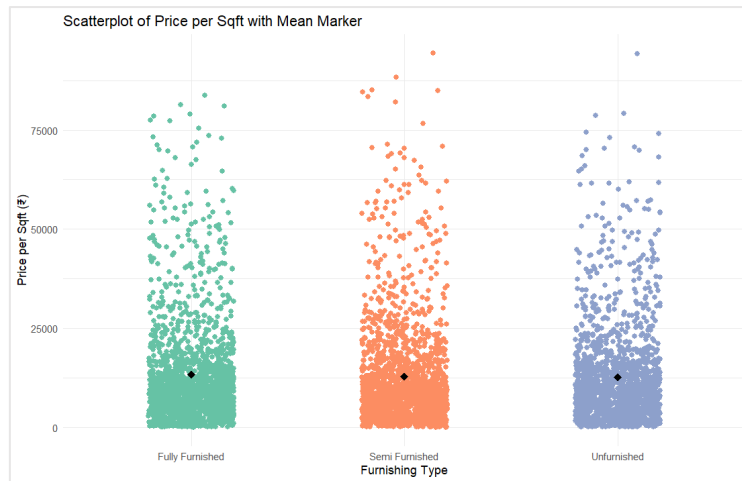


Fig 19: Price per square foot for different kinds of furnishing properties

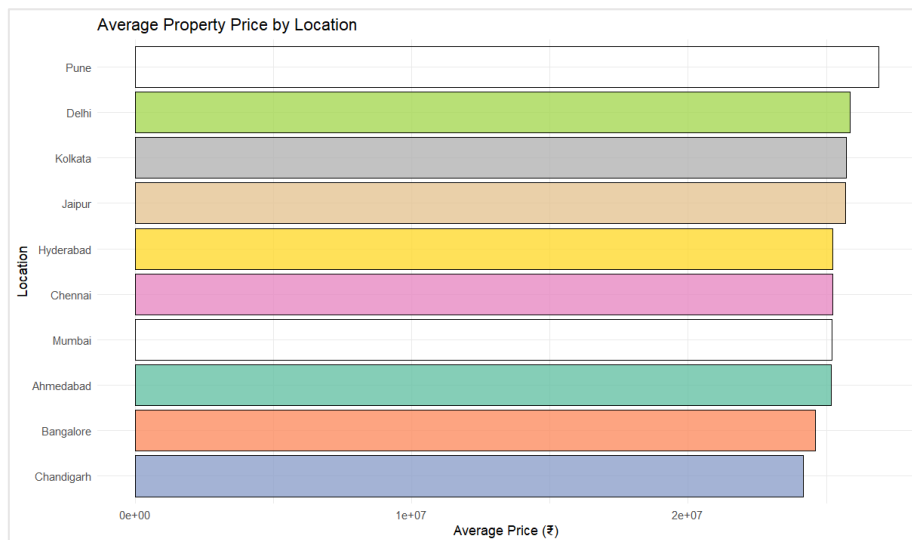


Fig 20: Average price of properties according to different locations

Results:

1. **Fig 1–3:** Apartments form the majority of listings, mostly with 2–3 bedrooms, while villas and independent houses typically have more rooms and larger layouts.
2. **Fig 4–5:** Most properties have 2–3 bathrooms; villas show higher bathroom counts, reflecting premium construction standards.
3. **Fig 6–7:** Semi-furnished units are most common overall, but villas are more likely to be fully furnished compared to apartments.
4. **Fig 8:** The top amenities include parking, security, garden, gym, and swimming pool—highlighting a focus on lifestyle and safety features.
5. **Fig 9–10:** Property sizes are right-skewed; villas and independent houses occupy larger plots, while apartments have smaller built-up areas.

6. **Fig 11–12:** Price data shows high variability; after log transformation, prices appear normally distributed, confirming initial right skewness.
7. **Fig 13:** Villas record the highest average price, followed by duplexes and independent houses; apartments are the most affordable.
8. **Fig 14–15:** Apartments are generally located closer to metro stations, whereas villas tend to be farther, showing an urban–suburban divide.
9. **Fig 17–19:** Apartments have higher price per square foot; fully furnished homes command premium rates across all property types.
10. **Fig 20:** Average property prices vary notably by location, with metro cities showing significantly higher values than smaller towns.

Conclusion:

The results directly address the problems outlined in the introduction—understanding how housing attributes, location, and infrastructure affect property prices in India’s urban real estate market. The findings imply that **location and connectivity**, particularly **proximity to metro stations**, play a decisive role in determining property value, supporting the idea that infrastructure greatly influences pricing. The dominance of **mid-sized, semi-furnished apartments** reflects a strong demand for affordable yet comfortable urban housing, aligning with the growing needs of middle-income buyers. Meanwhile, the premium pricing of **villas and fully furnished properties** indicates a distinct luxury segment driven by space, exclusivity, and amenities.

Overall, these results suggest that developers should focus on **strategic urban locations and moderate-size units with essential amenities** to meet mass-market demand, while policymakers and investors can leverage these insights to **plan infrastructure and pricing strategies** that balance affordability with growth in India’s evolving housing market.